Lab 6: bench checklist

Bulb and temperatures

Record the **bulb number** and use the same bulb for all runs. Read room temperature and the glass temperature at the marked equator point (emissivity 0.95 if adjustable). The reading is the **glass, not the vanes**. A **cooled start** is glass within 1 °C of room temperature with the rotor at rest. If the rotor does not start within 30 s at a gap, record “no rotation” and skip that movie.

Gaps and light

Use the course LED lamp at fixed power. Set the gap from its light-emitting front surface to the nearest glass surface at rotor height with spacer blocks or a ruler: **1.0, 1.5, 2.0, 3.0 and 4.0 cm, then repeat 1.0 cm after cooling**. Measure lux with the sensor at the rotor-centre position, bulb lifted off its mark, facing the lamp. Keep the background **25 cm behind the rear glass surface** and out of the strongest beam. Measure the lamp-face temperature once.

Camera

1280 × 1024 pixels; 1000 capture frames/s. Other settings are preset. Check a short movie, verify p over one full turn, and confirm that the recording holds at least seven passage marks plus 10 s after light-off.

Record

Glass temperature, then start recording before light-on. After seven or more passages, **switch the lamp off and keep recording**. Glass temperature again. Save 1cm_1.mp4, 1p5cm.mp4, 2cm.mp4, 3cm.mp4, 4cm.mp4 and 1cm_repeat.mp4 as H.264 with Text Overlay off. Note light-on and light-off frames and whether the rotor reversed. Wait for a cooled start before the next run.

Analyze

Mark_Lab6_Passage_Frames.m saves passage CSVs; run Run_Lab6_Radiometer.m for each. For 1cm_1, also mark the post-light-off passages as 1cm_1_coast_passages.csv and fit ln(rpm) versus time for T. Complete MIE402_Lab6_Run_Log.csv and the prefilled Lab6_Comparison_Log.csv (delete no-rotation rows), then run Compare_Lab6_Illumination_Runs.m with the same light-on window for all rows.

Keep

Raw movies, passage CSVs including the coast-down file, both logs and exported figures. The main guide lists the report figures and the required Q6 connection.